NJ Physics Professor Has the "Right Stuff"

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(Adapted by ReadWorks)



In 2005, Greg Olsen traveled to outer space. He was not a NASA astronaut who made a lot of money. He was a physics professor who had to pay $20 million for the trip. He spent ten days on a Russian rocket, taking part in a mission to the International Space Station.

Greg traveled to the space station with a crew of two other people. One of them was a Russian astronaut, and the other was an American astronaut. The main goal of their mission was to switch crews at the station. Another part of it was to replace some equipment at the station. The equipment would be used if anyone there ever had to make an emergency escape.

Going on this trip was a big achievement for Greg. He had been interested in space since he was a boy. But at the time he was born, nobody had ever gone into space. That was in 1945. Space exploration did not begin until the 1950s and '60s. In those years the United States and the Soviet Union were competing with each other to get into space first. In 1957, the first satellite made by people was sent into space from the Soviet Union. In 1962, the president of the United States said that Americans planned "to go to the moon" before 1970.

Now that space travel is a reality, Greg loves it. When asked why, he mentioned the astronauts who went into space with him. He said he loves space travel for the same reasons they do. What are those reasons? "[T]o be weightless, to see the awesome sight of Earth from space," he said. Fear was not a problem for him. He felt very confident about the kind of rocket he went on, known as a Soyuz rocket. It had safely taken other people into space, so he was not afraid.

Preparing for the Trip

Greg took charge of his own training for the trip into space. He went to space school, where he spent a lot of time in classrooms. He said it was "like being a college student again." He was expected to know his way around the rocket and the space station. He was also expected to help out every day on the space flight, but he was not given any big

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responsibilities.

Greg's training involved preparing his body for space as well. He spent time in flight simulators, which allowed him to experience what a zero gravity flight would be like. He said that the most difficult part of his physical training was a water-landing exercise. It prepared him for a type of landing called a "splash down." That is when a spacecraft uses a parachute to land on water. To practice this dangerous landing, Greg and the other people going on the trip wore wet suits and other items for protection. In two hours of practice he sweated so much that he lost three pounds.

Greg said the physical training was actually the easiest part of his preparation. What was the hardest part? Trying to learn Russian, he said. "I love Russians and the Russian culture… but I've never been good at languages since I was a young person." Still, he learned enough of the language to get along well with the Russian astronaut traveling with him. He also learned enough to get along well with the Russian people who helped with his training. They amazed him. "When I watched them operate the Soyuz spacecraft and the simulators, they seemed to know every nut and bolt on the vehicle," he said. "I just tried to soak up the knowledge."

Taking the Trip

The rocket Greg was on took off from a launch center near Russia. The launch went well. Greg said that lifting off from the ground was one of the best parts of his trip. He was also amazed at the sight of Earth going by in the rocket's window and by the feeling of floating around the International Space Station.

After he got to the station, he spoke on the radio. He spoke to his students in New Jersey. "Welcome to space," he said. "It's really nice here. It's nice and roomy. In some ways it's like camping out, because we have no running water, no sinks, and we kind of have to fend for ourselves for food," he said. He added that the astronauts with him made him feel welcome at the station.

Greg thanked a lot of people on the radio. He thanked professors, students, and old classmates at the universities where he had gone to school. He told his family and the people he worked with that he appreciated them.

Doing Studies in Space

Greg had planned to take a spectrometer into space with him. A spectrometer is a tool that uses light to help people collect information. It allows people to learn about objects in space. For example, an astronaut can use it to figure out what an object is made of. It can also be used to figure out an object's temperature, weight, speed, and direction.

Greg once built his own spectrometer and then started a company that built them. He was going to use one of his company's spectrometers on his trip. Then he learned that he

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couldn't take it out of the United States. He found other ways to study things in space, though.

He did three experiments to study the human body's reaction to zero gravity. He studied how being in space affects the lower back and inner ear. He also studied bacteria growth in zero gravity. Later he shared what he learned with the European Space Agency.

The Trip Home

On Greg's trip back to Earth there were some problems. They had to do with the pressure inside the spacecraft he was on. The problems were serious. They didn't lead to disaster, though. That was because the astronauts on board stayed calm and paid attention to the problems while returning to Earth. "At no time was there panic or alarm, or anything of that sort," Greg said. At one time during the return, he had to add more oxygen to the spacecraft. "We had practiced this many times during simulation practice, and I thought everyone handled it like pros," he said.

Dealing with these problems during the return to Earth was a test for the people on the spacecraft. It tested their skill, emotional strength, and mental ability. In the end, they landed safely back on Earth.

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